VoxShield SIH 26052 Technical Dossier

Team Name: Team@ Problem Statement ID: 26052

Technical objective

The technical objective is to build a real-time AI speech enhancement system for noisy defence communication. The system should suppress or reduce stationary, nonstationary and impulsive noise while preserving command speech. The intended deployment direction is edge hardware such as Raspberry Pi 5 connected to a controlled dual-microphone input path.

System definition

VoxShield consists of six major layers:

1. data and mixture generation layer
2. training and hard-case repair layer
3. streaming AI enhancement model
4. ONNX export and parity validation
5. edge runtime and benchmark layer
6. dashboard and evidence reporting layer

Audio input model

The model is designed around synchronized audio channels:

- primary channel: speech plus noise near the mouth or communication source
- reference channel: environmental noise reference

This is important because a single Bluetooth microphone or laptop microphone array does not provide a reliable raw reference channel. Physical deployment needs a dual-microphone hardware rig with stable timing.

Data sources used or planned

Source	Role
VCTK	clean speech
DNS Challenge	speech/noise challenge material
MAD dataset	military-like or hard acoustic material
ESC-50	environmental noise categories
AudioSet targeted subset	gunshot, explosion, vehicle, engine, siren, helicopter, aircraft
RIRS NOISES	room impulse responses and reverb
user recordings	personal clean/noisy examples such as fan and helicopter playback

Data ingestion status

The user has already ingested a large referenced catalog without copying audio. A later manifest included targeted AudioSet folders mapped into impulsive and nonstationary categories. The manifest counts showed more than 124,000 referenced files, with clean speech, environmental noise, impulsive noise, nonstationary noise, stationary noise and RIR categories.

This is useful because the deck can claim a real data pipeline, not a toy dataset.

Model and runtime

The current candidate model family is a dual-channel deep-filter model exported to ONNX. The model is lightweight enough for edge testing, around 1.1M parameters in reported slides. The model runs in a frame-based streaming style at 16 kHz with a 16 ms frame duration.

The runtime stack includes:

- Python training and evaluation tools
- ONNX export
- ONNX runtime inference
- Pi 5 benchmark
- live console or dashboard telemetry

Current strongest model candidates

Candidate	Strength	Limitation
v12 hard repair	strong metric candidate	older than v15
v14 realworld audioset	added more real-world targeted data	broad gate initially failed strict reliability
v15 hard failure repair	broad strict gate passed, strong export and metrics	military strict gate still fails reliability thresholds

Recommended deck candidate: v15 for recent narrative and broad gate proof. If comparing final numbers internally, keep v12 as a backup candidate if it outperforms in selected audits.

v15 export evidence

v15 ONNX export passed parity validation. Reported values:

- checkpoint: checkpoints_deep_filter_v15_hard_failure_repair/best.pt
- ONNX: artifacts/voxshield-v15-hard-failure-repair-final.onnx
- max output absolute error: about 1.82e-08
- max state absolute error: about 4.07e-06
- tolerance: 0.0002
- export status: passed

This proves the PyTorch model and ONNX export are numerically aligned enough for runtime deployment testing.

v15 broad audit

Reported v15 broad results:

- SNR enhanced: 30.48 dB
- SI-SNR enhanced: 30.64 dB
- STOI enhanced: 0.938

- PESQ enhanced: 3.94
- pause noise attenuation: 20.24 dB
- speech deletion rate: 0.0
- active level error: 0.10 dB
- strict SIH gate: passed

This should be used in the PPT as broad controlled-audit evidence.

v15 military audit

Reported v15 military results:

- SNR enhanced: 25.15 dB
- SI-SNR enhanced: 25.30 dB
- STOI enhanced: 0.915
- PESQ enhanced: 3.63
- pause noise attenuation: 24.58 dB
- speech deletion rate: 0.0
- active level error: 0.19 dB

The quality gate did not fully pass strict military reliability because STOI target reliability and joint target reliability missed thresholds. This should not be hidden in the technical dossier. It means the average is strong, but some hard clips still fail. That is a normal engineering finding and justifies the hard-case repair roadmap.

Pi 5 benchmark evidence

Reported Pi 5 verification:

- backend: ONNX runtime
- frame duration: 16 ms
- mean latency: about 5.33 ms
- p95 latency: about 5.49 ms
- p99 latency: about 6.05 ms
- full pipeline p99 callback: about 6.50 ms
- real-time factor: about 0.33
- deadline overruns: 0
- simulated callback drops: 0

This is one of the strongest project facts. Use it repeatedly, but do not overstate it as full headset validation.

Why live Bluetooth sounded bad

Bluetooth headsets commonly apply their own processing and use profiles that trade bandwidth, latency and microphone quality. Laptop and Bluetooth paths may:

- downsample or compress audio
- apply built-in noise suppression
- expose only mono audio
- fail to provide a clean reference channel
- add unpredictable latency
- prevent raw synchronized capture

Therefore, a bad Bluetooth demo does not automatically prove the model is useless. It proves the hardware path is wrong for validating physical cancellation.

Correct hardware architecture

Minimum next hardware rig:

- Raspberry Pi 5
- USB audio interface or multi-channel USB microphone device
- primary microphone near mouth
- reference microphone placed toward noise source
- wired headset or speaker output
- controlled noise playback source
- dashboard running locally or on same network

Avoid Bluetooth for proof demos.

Evaluation metrics

Metric	Meaning
SNR	energy-level noise reduction relative to clean target
SI-SNR	scale-invariant signal quality
STOI	intelligibility estimate
PESQ	perceptual quality estimate
speech deletion rate	checks whether speech was removed
active level error	checks whether speech loudness is preserved
latency	real-time feasibility

Technical risks and mitigations

Risk	Technical reason	Mitigation
Bad live demo with laptop/Bluetooth	no synchronized raw reference	use controlled replay and wired/USB audio
Model overfits synthetic mixtures	real noise differs	add field recordings and targeted hard-case repair
Speech deletion	aggressive suppression can remove speech	keep STOI and speech deletion gates
Impulse and rotor failures	hard transient/nonstationary noise	hard-failure dataset and loss weighting
Edge CPU load	real-time deadline risk	ONNX optimization and frame-level benchmark

Next build plan

Phase 1: freeze v15 as current demo candidate and keep v12 as backup.

Phase 2: create a controlled demo set:

- clean command
- vehicle noise mix
- rotor noise mix
- impulse noise mix
- fan/air noise mix
- one user-recorded scenario

Phase 3: build hardware rig:

- wired mic path
- synchronized dual input
- Pi 5 runtime
- dashboard

Phase 4: record real validation clips and rerun audit.

Technical conclusion

VoxShield is not yet a fully proven physical ANC product. It is a strong AI speech enhancement prototype with a validated edge-runtime path and a clear hardware validation roadmap. This is the correct, defensible technical story.